

Leveraged Return

In the previous calculations, we have assumed that the investor's position in an asset is equal to the total investment made by an investor using his or her own money. This section differs in that the investor creates a leveraged position.

There are two ways of creating a claim on asset returns that are greater than the investment of one's own money. First, an investor may trade futures contracts in which the money required to take a position may be as little as 10 percent of the notional value of the asset. In this case, the leveraged return, the return on the investor's own money, is 10 times the actual return of the underlying security. Both the gains and losses are amplified by a factor of 10.

Investors can also invest more than their own money by borrowing money to purchase the asset. This approach is easily done in stocks and bonds, and very common when investing in real estate. If half (50 percent) of the money invested is borrowed, then the gross return to the investor is doubled, but the interest to be paid on borrowed money must be deducted to calculate the net return.

Using borrowed capital, debt, the size of the leveraged position increases by the additional, borrowed capital. If the total investment return earned on the leveraged portfolio, R_p , exceeds the borrowing cost on debt, r_D , taking on leverage increases the return on the portfolio. Denoting the return on a leveraged portfolio as R_L , then the return can be calculated as follows:

$$R_L = \frac{\text{Portfolio return}}{\text{Portfolio equity}} = \frac{[R_p \times (V_E + V_B) - (V_B \times r_D)]}{V_E} = R_p + \frac{V_B}{V_E}(R_p - r_D), \quad (15)$$

where V_E is the equity of the portfolio and V_B is the debt or borrowed funds. If $R_p < r_D$ then leverage decreases R_L .

For example, for a EUR10 million equity portfolio that generates an 8 percent total investment return, R_p , over one year and is financed 30 percent with debt at 5 percent, then the leveraged return, R_L , is:

$$\begin{aligned} R_L &= R_p + \frac{V_B}{V_E}(R_p - r_D) = 8\% + \frac{\text{EUR3 million}}{\text{EUR7 million}}(8\% - 5\%) = 8\% + 0.43 \times 3\% \\ &= 9.29\%. \end{aligned}$$

EXAMPLE 15

Return Calculations

An analyst observes the following historic asset class geometric returns:

Exhibit 17: Asset Class Geometric Return

Asset Class	Geometric Return (%)
Equities	8.0
Corporate Bonds	6.5
Treasury bills	2.5
Inflation	2.1

1. The real rate of return for equities is *closest* to:

- A. 5.4 percent.
- B. 5.8 percent.